DPUTempomat

Simple cruise control system.

1. Inputs:

v	Current velocity [m/s].
Deactivate	If this input is set to 1, the cruise control changes from every state to the state Idle. If the input is set to value 0, the cruise control works normally. Default: 0
AcceleratorPedal	Current accelerator pedal position (Mockup)
BrakePedal	Current break pedal position (Mockup)
KeySet	Key to set the desired velocity. (Mockup)
KeyResume	Key to activate the cruise control with the last set desired speed. (Mockup)

2. Static parameters:

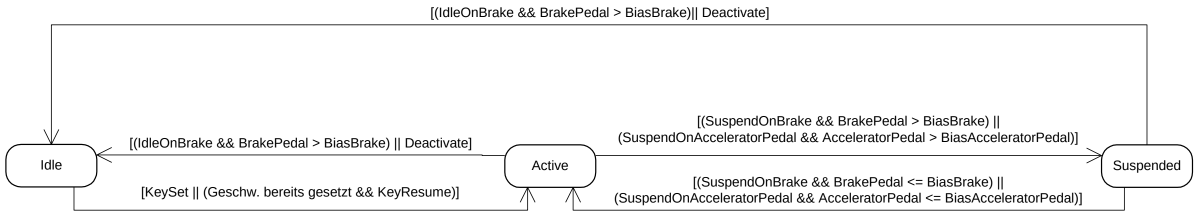
IdleOnBrake	Is this value not 0, then operating the brake results in the state Idle (see picture). Default: 1
SuspendOnBrake	Is this value not 0, then operating the brake results in the state Suspended (see picture). Default: 0
SuspendOnAcceleratorPedal	Is this value not 0, then operating the brake results in the state Suspended (see picture). Default: 1
BiasBrake	Threshold value from wich the operation of the brake pedal is recorded.
BiasAcceleratorPedal	Threshold values from wich the operation of the acceleration pedal is recorded.
kv1,kv2	Value of output kv at velocity *kxv1kmh* respectively *kxv2kmh* (for DPUDVControl). Between these values it is linearly interpolated.
kd1, kd2	Similar to kv1, kv2.
kxv1kmh, kxv2kmh	see above

3. Outputs:

State	State. 0 = Idle, 1 = Active, 2 = Suspended (see picture)
ActivateDVControl	Activate velocity control (to DPUDVControl).
vTargetOut	Current desired velocity (to DPUDVControl).
kv, kd	Controller parameters (to DPUDVControl).

4. Functional principle:

The cruise control works according to the following state diagram:



The interconnection of the DPU with the Mockup, the DPUDVControl and the DPUAxControl shows this configuration excerpt:

```
Connections =
{
      VDyn.v <-> Tempomat.v,
      Mockup.BrakePedal <-> Tempomat.BrakePedal,
      Mockup.AcceleratorPedal <-> Tempomat.AcceleratorPedal,

      # => DVControl
      Tempomat.ActivateDVControl <-> DVControl.Activate,
      Tempomat.vTargetOut <-> DVControl.vTarget,
      Tempomat.kv <-> DVControl.kv,
      Tempomat.kd <-> DVControl.kd,
      VDyn.v <-> DVControl.v,

      # => AxControl
      DVControl.ActivateAxControl <-> AxControl.Activate,
      DVControl.axTarget <-> AxControl.axTarget,
      VDyn.v <-> AxControl.v,
      Mockup.AcceleratorPedal <-> AxControl.GasPedalIn,
      Mockup.BrakePedal <-> AxControl.BrakeIn,

      # => Driving dynamics
      AxControl.GasPedalOut <-> VDyn.AcceleratorPedal,
      AxControl.BrakeOut <-> VDyn.BrakePedal
};
```